

Milestone I

Project Plan

Title

Design of Lightweight Broad-Band Microwave Absorbers

Sponsor

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1. Project Background

Summary of Need

The use of lightweight broad-band microwave absorbers in defense applications play a key role in reducing the electronic signature of military bases, vehicles, and personnel, thereby enhancing stealth capabilities, survivability, electronic warfare, and the protection of sensitive equipment. This project aims to adapt existing python code, which solves Maxwell's equations for multi-layer films at optical frequencies, to work with radar frequencies.

The existing python code currently solves for a three-layer system surrounded by air. Adapting the python code to solve for different surrounding materials, such as perfect electrical conducting (PEC) material, allows for broader use of the existing code. In addition, the implementation of machine learning to the existing python code will optimize the speed and solving capabilities of the code, producing the best result using the given materials.

Industry Served

Radiation absorbing materials (RAM) have been a target of research for the Department of Defense because of increasing threats to the national security of the United States and its allies. These materials are under research for their stealth capabilities for absorbing radar frequencies. The coatings not only need to exhibit a low reflection coefficient over a wide frequency range, but also need to be lightweight and thin. Designing such materials is challenging though, absorbing too many radar frequencies is not advantageous because it looks suspicious, but absorbing too little will allow what is hidden to be seen.

Multilayered coatings are used today in the absorption of radar frequencies. Usually, the multilayered coating is surrounded by just air or by air on one side and PEC on the other. Fig. 1.01 depicts a multilayered system with N_m different materials with frequency dependent permittivity $\epsilon_i(f)$ and permeabilities $\mu_i(f)$ backed by a perfectly conducting material.

Excerpt from – “Design of Lightweight, Broad-Band Microwave Absorbers Using Genetic Algorithms”

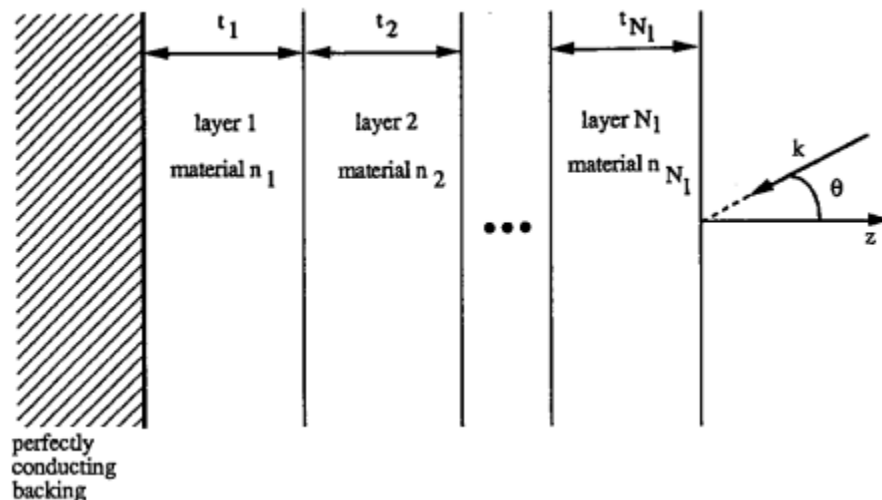


Figure 1.01: Multilayered RAM Coating with Perfectly Conducting Backing.
Adapted from ieeexplore.ieee.org

Modern RAMs can be classified into passive and active materials. Passive RAMs, which include dielectric, magnetic, and composite materials, absorb radiation without requiring an external power source. Active RAMs, however, rely on adaptive technologies that can change their absorption properties in response to environmental conditions.

Currently approaches include the use of conductive polymers, carbon-based materials, and ferrite coatings. However, these materials are limited by thickness, weight, and effectiveness across a broad frequency range.

According to industry research, the global market for electromagnetic shielding materials is projected to grow from \$7.14 billion in 2020 to \$11.78 billion by 2027, driven by increased demand in aerospace and defense industries for more effective RAMs. Current research focuses on optimizing the material layers to achieve higher absorption at lower thickness, such as using nanomaterials, which provide increased surface area for better attenuation.

A leading study by Zhang et al. (2021) demonstrated that graphene-based multilayered RAMS could absorb 99% of incoming radiation at specific frequencies, significantly outperforming traditional metal-based shields.

The anticipated outcome of this project is a multilayered RAM that can be applied to military aircraft, spacecraft, and communication satellites. This material will offer superior absorption at reduced thickness and weight compared to current technologies, addressing the growing demand for lightweight, high-performance shielding.

Today, RAMs are often bulky, requiring thick layers to achieve effective absorption. The use of single-layered absorbers limits performance over a wide frequency range, making them unsuitable for multifunctional applications. Ferrite-based materials, for instance, are effective at lower frequencies but fail at higher ones. Similarly, traditional metal-based shields are heavy and can only cover a limited frequency range.

The limitations of current RAMs include:

- High weight: Restricts usage in aviation and space applications.
- Frequency specificity: Most RAMs are effective only over narrow frequency bands.
- Cost: Advanced materials like ferrites and conductive polymers can be expensive and difficult to manufacture.

References

1. Balanis, C. A., "Advanced Engineering Electromagnetics," Wiley, 2021.
2. Robinson, M. T., et al., "Multilayered Materials for Electromagnetic Interference Shielding," *Journal of Applied Physics*, vol. 132, no. 6, 2022.
3. Zhang, W., et al., "Graphene-Based Absorbing Materials for Electromagnetic Shielding," *Nature Nanotechnology*, 2021.
4. Prasad, N. B., and Singh, K., "Development of Lightweight Stealth Materials," *Journal of Aerospace Materials*, 2020.
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6. Michielssen, E., et al. "Design of lightweight, broad-band microwave absorbers using genetic algorithms." *IEEE Transactions on Microwave Theory and Techniques*, vol. 41, no. 6, 1993, pp. 1024–1031, <https://doi.org/10.1109/22.238519>.

End User

The primary end users are military aircraft manufacturers, space agencies, and defense contractors. These organizations require RAMs to reduce the radar cross-sections of aircraft, unmanned aerial vehicles (UAVs), and satellites. The stealth technology community will benefit from lighter and more effective materials that reduce radar detectability while maintaining structural integrity.

The novel approach in this project involves the use of multilayered materials that optimize absorption across a broad frequency range. Current RAMs are limited by their single-layer design, which can only cover narrow frequency bands or require thick, heavy layers for effectiveness.

This approach will be successful due to its ability to address the weight and frequency limitation issues in current materials, particularly in aerospace applications where weight reduction is critical.

Python code will also be adapted in this approach to include machine learning that will optimize the problem-solving capabilities and variety of applications that can be solved.

The community most impacted will be military personnel. For defense, stealth technology advancements are crucial in maintaining strategic advantages.

If successful, this project will lead to significant advances in military and defense applications by achieving lighter, more effective RAMs, enhancing stealth capabilities and reducing operational costs.

Risks involved with this project include the manufacturability of the material and the implementation of the machine learning algorithm. Since Python is being used, the material will not actually be manufactured, just simulated using the code, this may bring problems with manufacturing later because the material may not be able to be feasibly manufactured. The implementation of the machine learning algorithm is also a challenge because of the complexity of the task.

If successful the payoffs will be better performance of the RAM, being able to absorb more radiation and being thinner therefore reducing the weight of the material.

The end user will directly experience the practical benefits, with lighter, more efficient, and adaptable RAMs, offering improved functionality in critical technological and defense applications.

Motivation

The primary motivation for this project is to meet the growing need for advanced stealth technologies and electromagnetic shielding in military applications. Modern warfare increasingly relies on technological superiority, including the ability to evade detection and protect sensitive equipment from electromagnetic interference (EMI). Traditional methods for developing radiation-absorbing materials (RAMs) struggle to keep up with evolving demands for lighter, more efficient, and broader-spectrum materials. This project aims to leverage machine learning to optimize the design of multilayered RAMs specifically for military use.

In this context, machine learning offers a way to enhance the development process by rapidly analyzing large datasets of material properties, radar absorption tests, and environmental conditions. The Python-based machine learning models will simulate and predict the most effective RAM configurations, speeding up the design process and delivering superior materials to the field. This approach aligns with the defense sector's ongoing push toward data-driven design and AI-powered innovation in materials engineering.

By optimizing RAM design for military platforms, this project directly contributes to improving the survivability of military assets and ensuring they remain undetectable in increasingly contested operational theaters.

Significance and Impact

The development of multilayered radiation-absorbing materials (RAMs) for military applications has the potential to significantly enhance the effectiveness of stealth technology, improve electromagnetic interference (EMI) shielding, and provide greater operational security in modern warfare. The materials developed through this project could revolutionize the aerospace, defense, and electronic warfare sectors by creating more efficient, lightweight, and adaptable RAMs for vehicles, aircraft, drones, and other critical military assets.

Quang Nguyen works at the Army Research Lab tasked with researching these multilayered materials. His role not only involves the funding of cutting-edge research but also collaboration with military end-users to ensure that the materials produced meet the demands of real-world operation environments.

Societal issues include national security, reduction of collateral damage and technological leadership.

The enhanced stealth capabilities provided by advanced RAMs directly contribute to national defense. In an increasingly volatile geopolitical landscape, having the technological advantage to remain undetectable by adversaries' radar systems offers critical strategic and operational benefits.

Stealth technologies, enhanced by more efficient RAMs, can reduce the need for broad-spectrum, high-impact military engagements. More accurate and undetectable operations can be conducted with precision, potentially minimizing civilian casualties and reducing the risk of collateral damage in combat zones.

References

1. Robinson, M. T., et al., "Multilayered Materials for Electromagnetic Interference Shielding," *Journal of Applied Physics*, vol. 132, no. 6, 2022.
2. Prasad, N. B., and Singh, K., "Development of Lightweight Stealth Materials," *Journal of Aerospace Materials*, 2020.
3. Zhang, W., et al., "Graphene-Based Absorbing Materials for Electromagnetic Shielding," *Nature Nanotechnology*, 2021.

Project Continuation

This project is a first-year project, but Python code was provided by the Army Research Lab to adapt it for implementation of a machine learning algorithm. The continuation of this project would be for

teams to create a physical material that replicates the properties used by the code for the efficient absorption of radiation.

2. Specifications and Requirements

Project Design Constraints

The primary design constraint for this project is simulating a multilayered material system for radiation absorption using Python. While the simulation offers an optimal solution for the material's performance, it introduces limitations when translating the results into real-world applications. The feasibility of manufacturing the proposed material remains uncertain.

Additionally, the code does not account for the cost of production, which means that while the material may be highly effective at absorbing radiation, it could be prohibitively expensive to produce. Furthermore, the environmental impact of sourcing and mining the raw materials required for the design is not considered, which could pose significant risks.

Requirements & Specifications

The design must adhere to three key requirements:

1. Utilize the 16 materials provided and their respective properties for the final material design.
2. Modify the existing Python code to optimize the radiation absorption of the multilayered material.
3. Develop a machine learning algorithm to enhance both the radiation absorption performance and the computational efficiency of the code.

To achieve this, the machine learning algorithm must be seamlessly integrated into the existing Python framework. Additionally, the Python code should be vectorized to eliminate 'for' loops, improving efficiency. Equations must also be implemented to convert the given material properties into the format required by the Python code for accurate simulation.

Summary Mind Map or Quad Chart



Figure 2.01: Miro Mind Map

Summary of Need

The use of lightweight broad-band microwave absorbers in defense applications play a key role in reducing the electronic signature of military bases, vehicles, and personnel, thereby enhancing stealth capabilities, survivability, electronic warfare, and the protection of sensitive equipment. This project aims to adapt existing python code, which solves Maxwell's equations for multi-layer films at optical frequencies, to work with radar frequencies.

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3. Project Objectives

Project Goal

Create a radiation absorbent material (RAM) optimization model using Python to advance stealth and survivability technology for the US military. This neural network must optimize material selection and automate the process for future development of the technology.

Objectives

*Objective 1: **Make improvements to Python and MATLAB code.** The code provided to us is rough and unfinished. We are tasked with making it more readable and commented for future development. Commenting each line will also improve the teams understanding of the topics and*

code. Additionally, we are to vectorize the MATLAB code by replacing the for loops. This is known to improve the performance and efficacy of the script. Two finished codes will be made for Python and MATLAB to be presented to the sponsors for review.

***Objective 2: Add material specifications to Python code.** We must adapt 16 fictitious materials that were given in our sponsors research paper. This will allow the code to pull from a wider database for designing the layer selection. We must also *replace back layer with perfectly electrical conducting material as per the sponsors requirement. To ensure this is all done properly and reliably, we will implement test functions to validate the new commands. We will show the sponsors our progress for assurance that our work is correct and aligns with their expectations.**

***Objective 3: Develop AI algorithm/machine learning to optimize code.** Our largest task is to design and implement 3 neural network algorithms to find the most efficient and highest performance model. The three models are: linear regression, swarm, genetic. The implementation of this model will allow future development of the code through automation and up-to-date design methodologies. Finally, we will train the model using existing data. We will present our progress on all three models to our sponsors along the entire development for review and comments.*

Endpoints – Deliverables and Metrics for Success

The anticipated project outcomes include an enhanced version of the original Python and MATLAB codes. The updated Python code will leverage machine learning techniques to retain memory from previously obtained results, facilitating the design of an optimized microwave-absorbing material composed of three distinct components. Meanwhile, the MATLAB code will utilize vectorization instead of traditional for loops, improving both readability and performance in the search for the optimal material design.

Additionally, unit tests will be incorporated into both the MATLAB and Python codes to ensure the accuracy of the results. To further validate these results, hand calculations of microwave absorption can be performed and compared against the outputs generated by the respective codes.

3. Project Planning

Resources

Our required resources for this project include Python and MATLAB. There are also several research papers that our sponsors provided, or we found that are fundamentally important to understanding the objectives of our project. We do not require hardware or physical resources at this time.

Anticipated Design Iterations

Our two largest goals are to add 16 additional resources into the Python code database, and to implement a machine learning algorithm. We anticipate the greatest area of iteration to be in developing the algorithms due to our lack of experience and attempting to achieve higher efficiency.

Within our Gantt chart we have outlined our process for iterating the machine learning algorithms. We are to attempt three separate algorithms. For each one, we have an initial build phase, adding test functions to ensure they are working as intended, comparing results to other algorithms, and finally presenting results to our sponsors. This entire process is designed to be iterative as it is predicated on continuously reviewing and improving our algorithms. We have strict deadlines for all phases to keep us on schedule for a timely delivery.

Support from Project Sponsor

We have meetings set up for every Thursday with our sponsors where we review progress, go over any challenges, and seek assistance on future goals. Additionally, we have weekly update email correspondence with our sponsors every Monday with a short summary of what we are actively working on to keep them apprised of our timeline. This ensures they are able to offer support at any point along our effort.

Support from Swanson Center for Product Innovation

At this stage, we do not anticipate needing to use the Swanson Center for Product Innovation. Our project is purely software based. With our current goals and sponsor recommendations, we are focusing on advancing the Python and MATLAB script.

Support from Subject Matter Experts

Due to the unique nature of this challenge, it necessitates substantial research and communication with subject matter experts – especially in regard to implementing neural networks. As such, two initial meetings have been set up. We are meeting with Dr. Bajaj, a MEMS professor that is proficient in Python and adapting AI algorithms. Secondly, we are meeting with Dr. Dallal, the lead machine learning research professor in the COE department. We will seek initial guidance on how to best approach developing a neural network for our specific need. We will then hopefully have brief, bi-weekly, follow-up meetings to ensure our work is correct and future goals are established.

Potential Risks and Anticipated Failures

One major risk presented with this project is misuse of time and getting bogged down in one objective for too long. We plan to address this obstacle by staying true to our timeline and Gantt chart. We will hold each other accountable and perform weekly check-ins to ensure we are maintaining our goals.

A second obstacle is the design and implantation of the neural network algorithms. This is an aspect of engineering that we as mechanical engineers have not had too much exposure to. Consequentially, we have, and continue to do, substantial research on how to develop AI algorithms. Furthermore, we have set up meetings with subject matter experts to assist us in any roadblock we are sure to face. To address this, we plan to continue doing what we have done in searching for help. We will also continue reviewing our work with our sponsors to ensure we are developing their ideas as we are expected to.

Milestones

Our Milestones are listed below:

September: Initial Planning and Research

1. Project Kickoff Meeting
 - o Define project scope, objectives, and deliverables
 - o Establish roles and responsibilities for each team member
2. Gathering Requirements Details and Analysis
 - o Document functional and non-functional requirements
 - o Identify data sources, papers, and any necessary tools or libraries in Python/MATLAB
3. Technical Research and Feasibility Study
 - o Explore existing neural network frameworks through provided research papers
4. Environment Setup
 - o Set up development environment (Visual Studio)
 - o Ensure all team members have access to necessary resources, software, and communication channels
5. Initial Prototype Development
 - o Develop a simple prototype to test basic functionality
 - o Meet with subject matter experts to explore algorithm implementation

October: Development and Testing

1. Model Design and Architecture
 - o Finalize the neural network architecture based on project requirements
 - o Create diagrams/documentation of three competing model structures
2. Implementation of Neural Network
 - o Begin coding the 3 neural network models in tandem
 - o Implement training and validation routines
3. Data Preparation and Augmentation
 - o Prepare the dataset for training
 - o Apply data augmentation techniques if necessary
4. Initial Training Phase
 - o Train the model on a subset of data
 - o Monitor performance metrics on three models
5. Mid-Project Review
 - o Assess progress against initial timelines, create new goals, gather feedback
 - o Identify and address any challenges.

November: Refinement, Deployment, Presentation

1. Model Evaluation and Fine-Tuning
 - o Evaluate the models using validation data
2. Final Testing
 - o Conduct thorough testing, choose best performing neural network
 - o Ensure model generalization and robustness
3. Documentation and Code Review
 - o Complete documentation for the codebase and model
 - o Conduct code reviews with the team and sponsors
4. Deployment Preparation
 - o Prepare the model for deployment
 - o Develop any necessary APIs or interfaces within the Python script
5. Final Review and Project Wrap-Up

- o Create poster and supplemental presentation items.
- o Gather feedback and document lessons learned

Budget

Our budget has been declared to be \$2500. We do not anticipate any costs arising for this project. Our sponsors suggested the possibility of traveling to a conference in Maryland using our allotted funds if we have a successful end result.

Timeline

See pittmems.teamwork.com for view of project gantt chart

4. Project Execution

Analysis & Design

The anticipated analysis will integrate concepts from material science, nuclear engineering, and mechanical measurements. Given the nature of our project, we will employ techniques from our MEMS courses to evaluate various aspects of our work. To identify an optimized microwave absorbing material, we will leverage our understanding of materials science to analyze the electromagnetic properties of different materials, focusing on their interactions with microwave frequencies. By interpreting graphs generated from our Python code, we can refine our design for optimal performance.

We will also incorporate nuclear engineering principles, such as radiation types and behaviors, to gain a foundational understanding of the problem at hand. A key aspect of this understanding will involve documenting and commenting on the existing Python and MATLAB codes we received, ensuring that all team members comprehend each command and its relevance to microwave absorption.

Additionally, we will apply mechanical measurement techniques throughout the project, particularly in calculating uncertainties associated with each data set generated by our codes. This will involve assessing the uncertainty in microwave absorption for each material derived from the Python code and comparing these results to the benchmarks established by the MATLAB code.

Verification

To verify our project endpoints and Python results, we will implement unit tests to validate any new commands we introduce to the code. These tests will help ensure that our code is error-free and will allow us to trace any issues as they arise. Since we are primarily adding machine learning components to the Python code, we can benchmark our optimized results by comparing them with the outcomes from the existing MATLAB code. While we don't expect the results to match perfectly, they should be closely aligned.

5. Project Organization

Team Meetings

As a team, we meet three times a week on Microsoft Teams for an hour each session. Our planner, Abegale, diligently takes notes during each meeting and stores them in the TeamWork files section for easy access for all team members. Important data from our sponsor's research, along with our MATLAB and Python results, is compiled in a shared Google Document that is updated by all team members whenever new data is added. This document is also stored in the TeamWork files category, ensuring that everyone has access to the latest information.

Work Space

Our primary workspace is the Benedum Library, where we collaborate effectively on enhancing our MATLAB and Python codes in an environment that fosters focus. As our project involves improving two existing codes, we will store our work on our laptops for easy access. Whenever we make updates to the code, we share the latest versions with all team members via TeamWork, along with detailed comments documenting those changes. The nature of our project allows us to work without any special conditions, making the library an ideal setting for our teamwork.

Sponsor Meetings

Our group meets weekly with our project sponsor for approximately an hour. These meetings are held on Microsoft Teams, ensuring easy attendance for everyone involved. We have obtained consent to record all sessions, allowing us to reference them later, and we store these recordings in our team Discord for easy access. Additionally, we communicate with our sponsor via email throughout the week whenever questions arise.

Sharing, Distribution and Archival of Project Data

All project data, including updated versions of our MATLAB and Python code, as well as meeting minutes, are stored in the TeamWork files section. This ensures that all team members have easy access to the recorded information and can quickly locate relevant project details. We also utilize email to communicate and share important updates with our sponsor.

Return of Equipment

Our project sponsor has provided us with a forked version of the existing MATLAB and Python codes to ensure that we don't inadvertently alter the core structure, which could compromise their previous work. At the conclusion of our project, we will deliver the finalized versions of each code to our sponsors, Dr. Leu and Mr. Nguyen. Our resource manager, Joseph, will be responsible for overseeing this handover.

6. Project Outcomes

Community

Our project will have a significant impact on the United States Department of Defense and its allies. Our research aims to develop an optimized microwave-absorbing stealth coating specifically designed for U.S. military vehicles, buildings, and aircraft. This innovative solution has the potential to enhance the effectiveness and protection of critical defense assets.

Educational

We anticipate developing both soft and technical skills throughout this project. This experience will enhance our teamwork abilities, allowing us to navigate diverse perspectives and strengthen our conflict resolution skills as we work towards consensus. Additionally, managing our project timeline will hone our time management skills, helping us meet deadlines while balancing individual responsibilities. Moreover, we are gaining valuable experience in professional networking, further enriching our collaborative journey.

The technical skills we anticipate developing include improving our MATLAB and Python coding capabilities as well as deepening our understanding of nuclear concepts. In MATLAB, we are focusing on vectorizing for loops to enhance code readability and improve performance. Additionally, we will learn to implement a machine learning algorithm in Python that can store data from previous runs, resulting in increased efficiency and faster execution. Throughout this process, we will also explore various nuclear concepts, including the electromagnetic properties of materials and their relationship to microwave absorption.